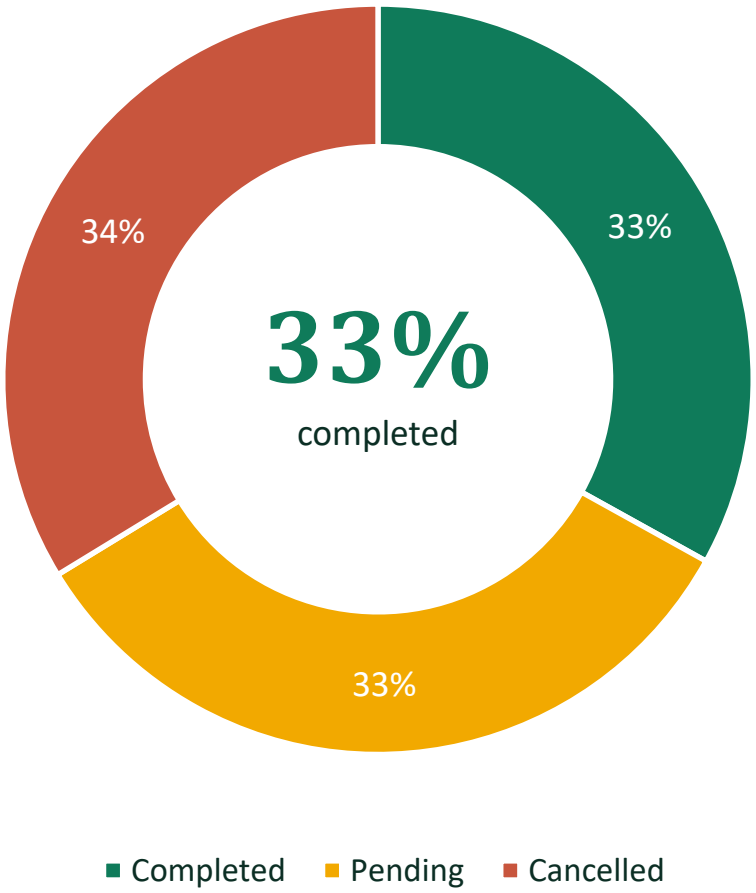
ShopSmart Retail: what 50,000 orders reveal



Exploratory Data Analysis | Data Science Track, Week 1

AnalystLab Africa Experience Lab | Presented by Charity Nduati

Only 1 in 3 orders is Completed



₦2.25bn

Realised revenue (Completed orders)

₦6.82bn

Booked order value (all statuses)

₦4.6bn

Stuck in Pending or Cancelled orders

Key performance indicators

16,529

Completed orders

4,795

Customers who bought

₦135,933

Average order value

4.83%

Return rate

₦3.17bn

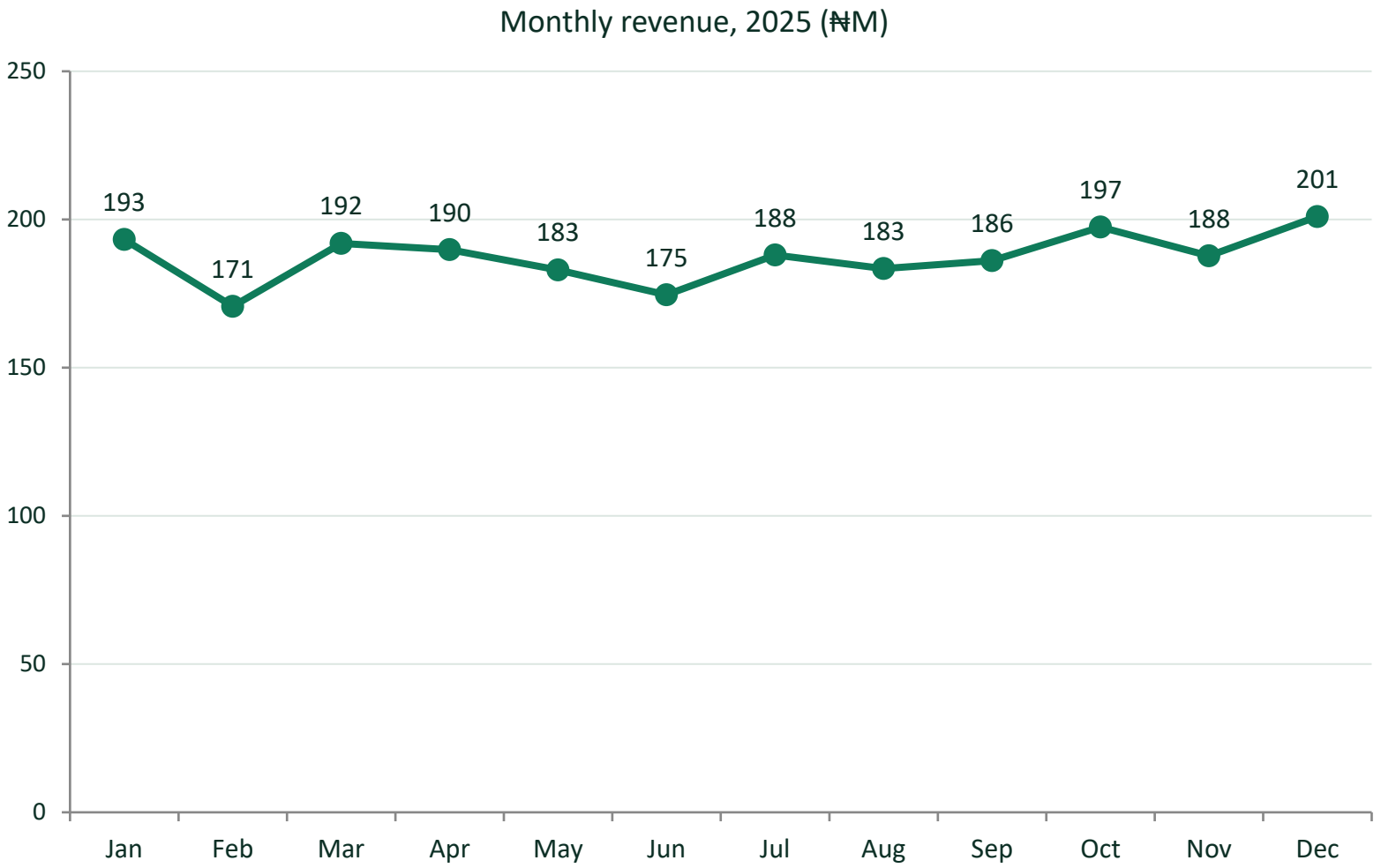
Inventory value

25 | 500

Stores | Products

Revenue = quantity × unit price × (1 − discount %), Completed orders only. Inventory value = stock × cost price.

Sales are steady all year, peaking in December



₦201M

December (peak)

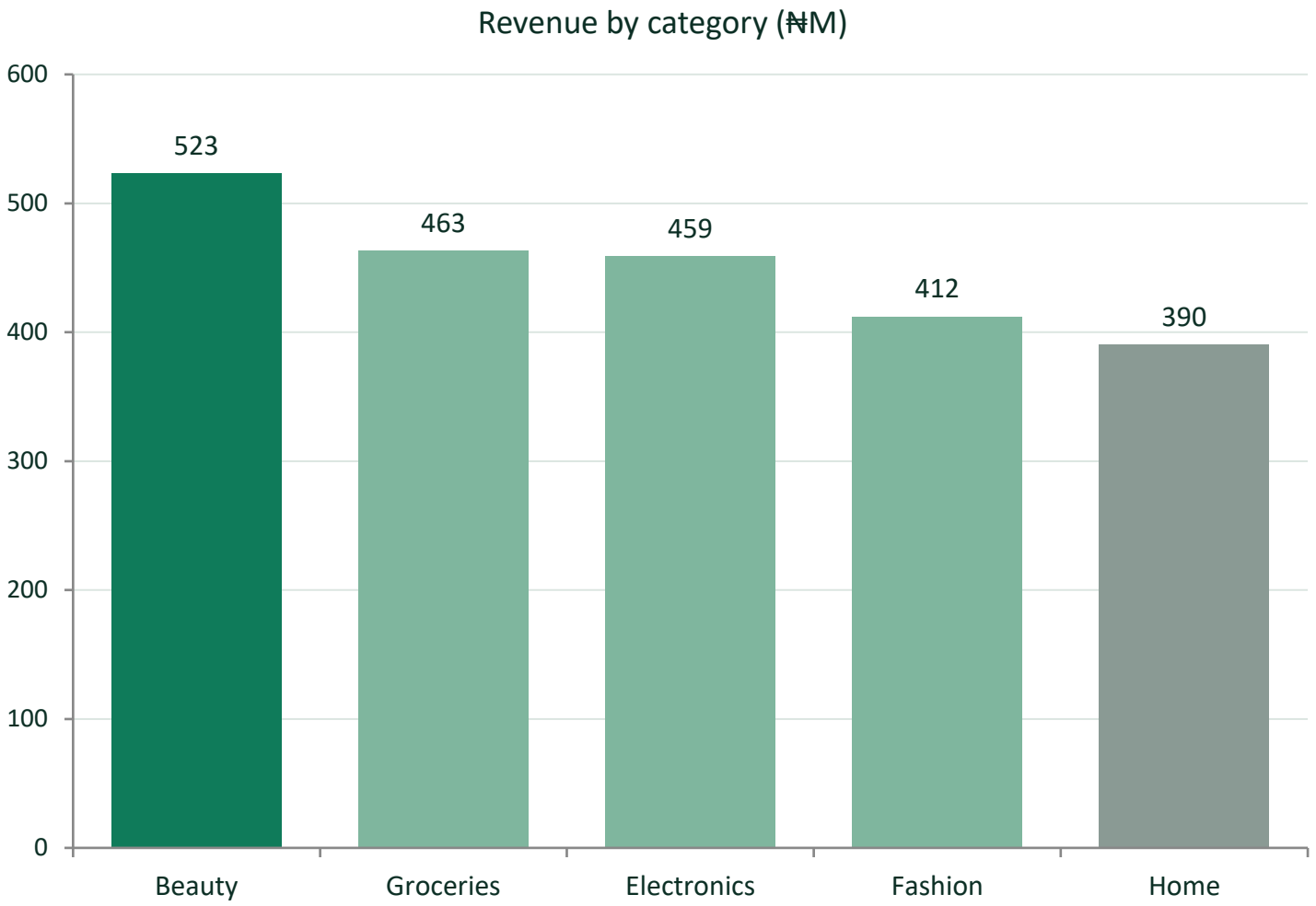
₦171M

February (low)

±17%

Spread: little seasonality

Beauty leads; Fashion is growing fastest



Beauty: ₦523M

23.3% of all revenue

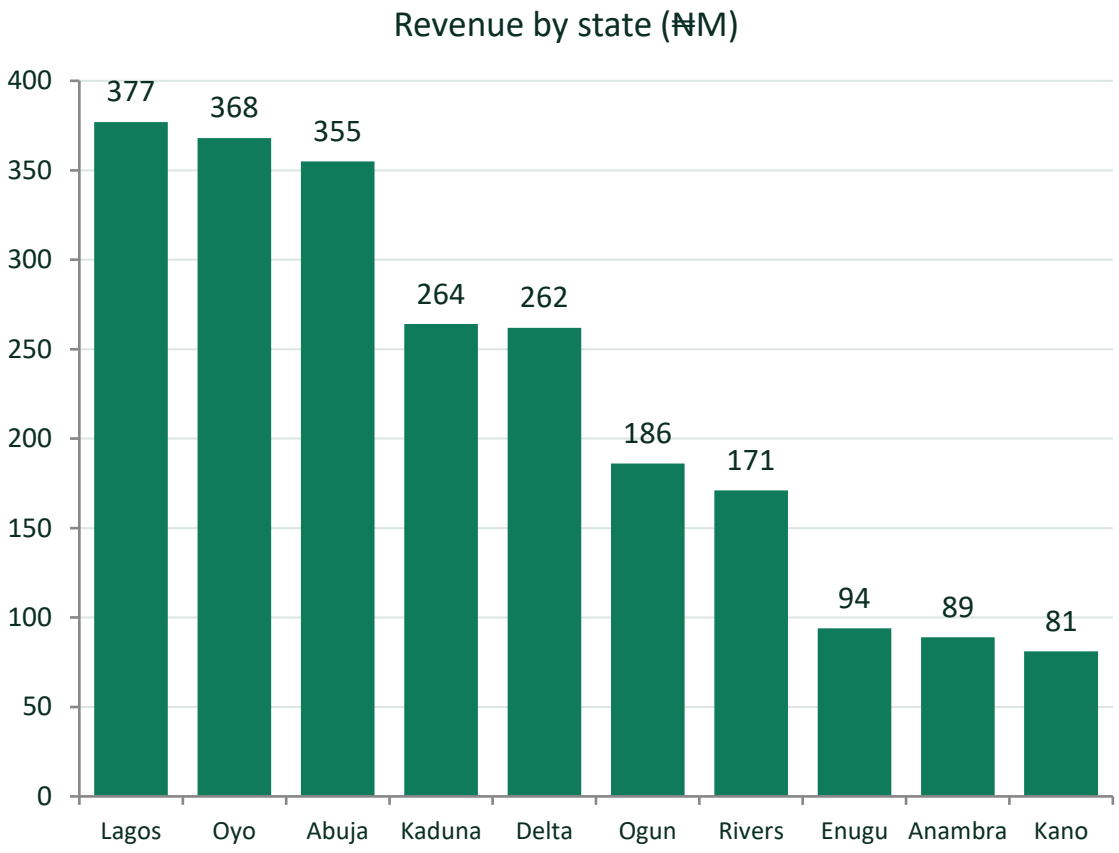
Fashion: +6.2%

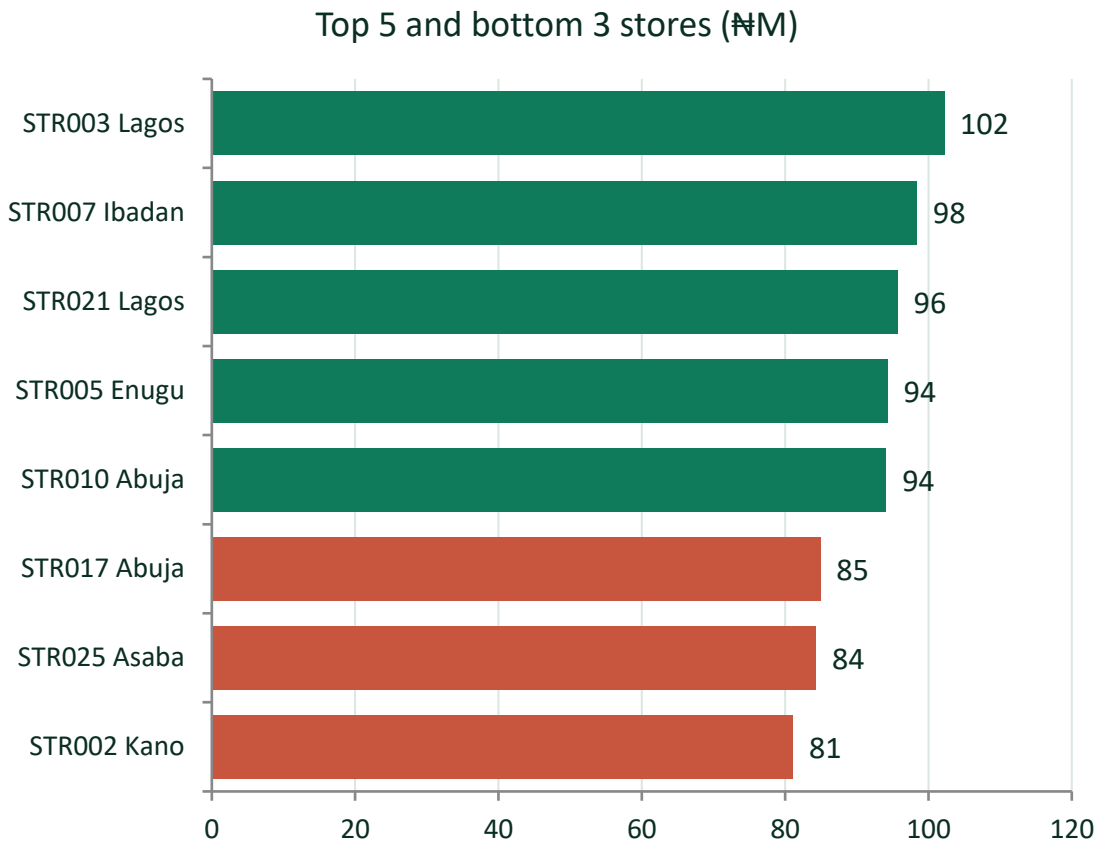
H2 vs H1, and +10% Q4 vs Q1

Home: ₦390M

Lowest revenue, slowest growth

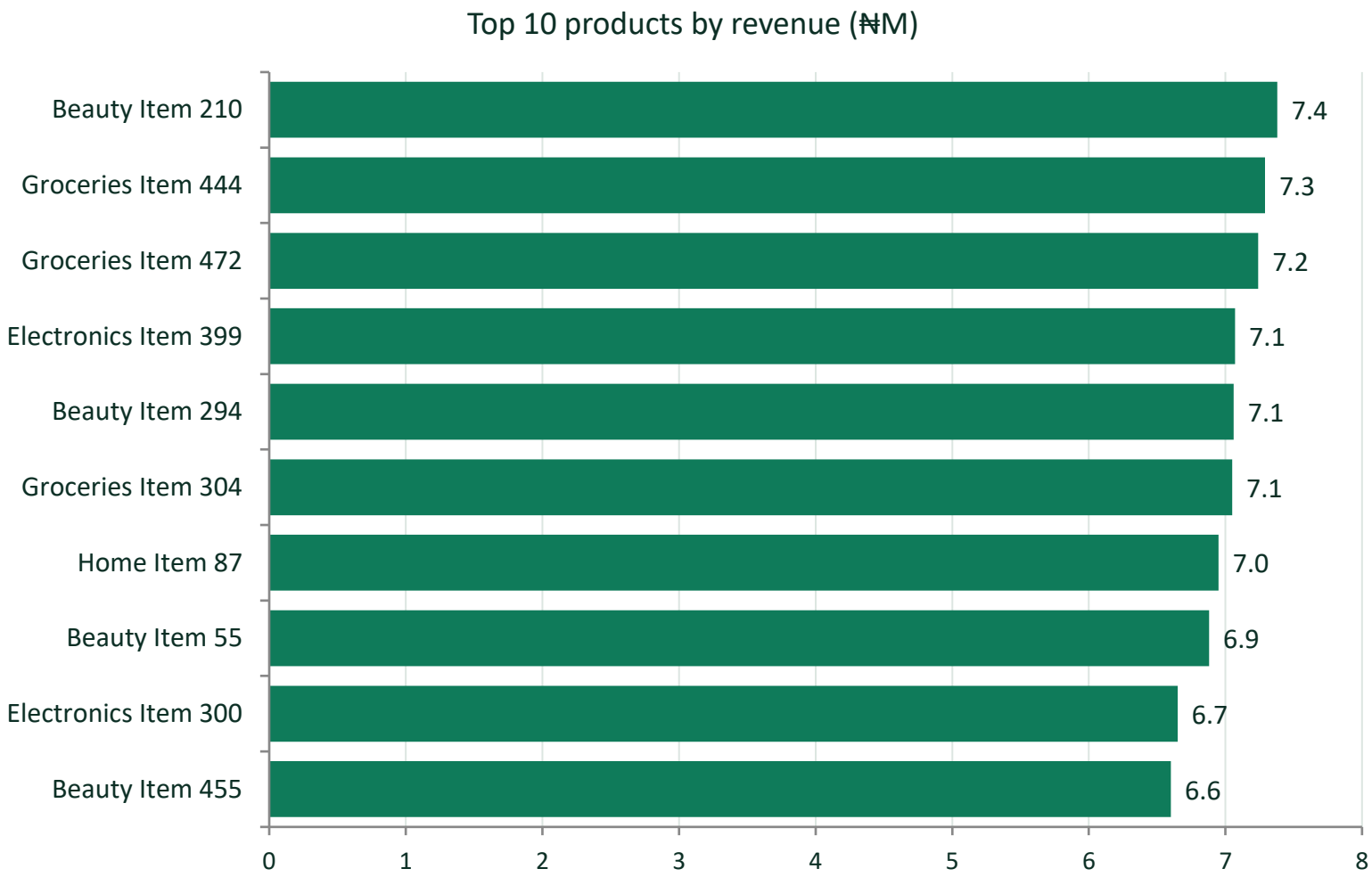
Stores and states perform within a narrow band





Best store: STR003 Lagos (₦102M). Weakest: STR002 Kano (₦81M). Per store, Lagos and Enugu lead (₦94M). Always group by Store_ID: store names repeat.

No single product dominates



₦7.4M

Top product: Beauty Item 210

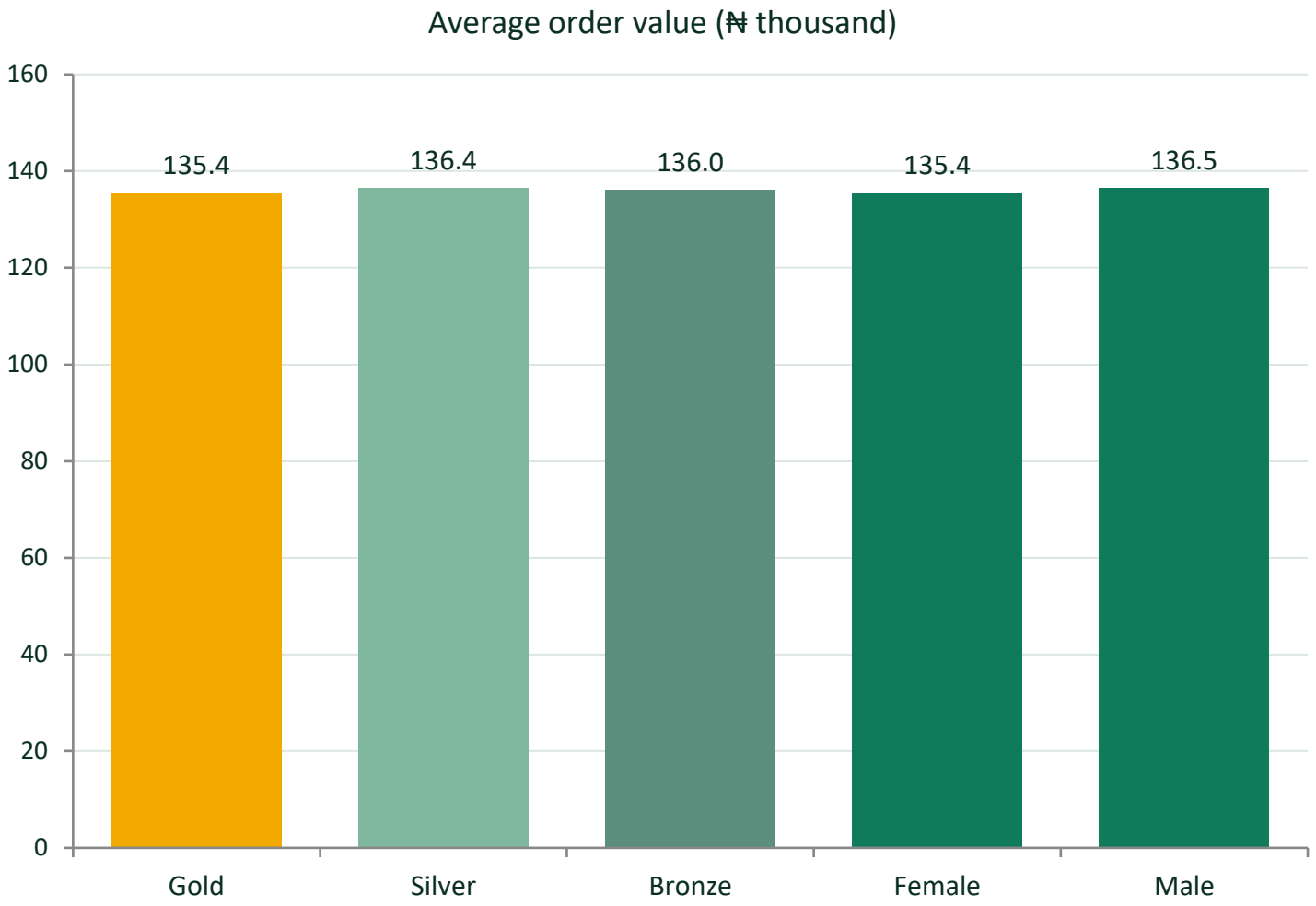
~5×

Top vs weakest product

0.3%

Share held by the top product

Loyalty tiers do not change what customers spend



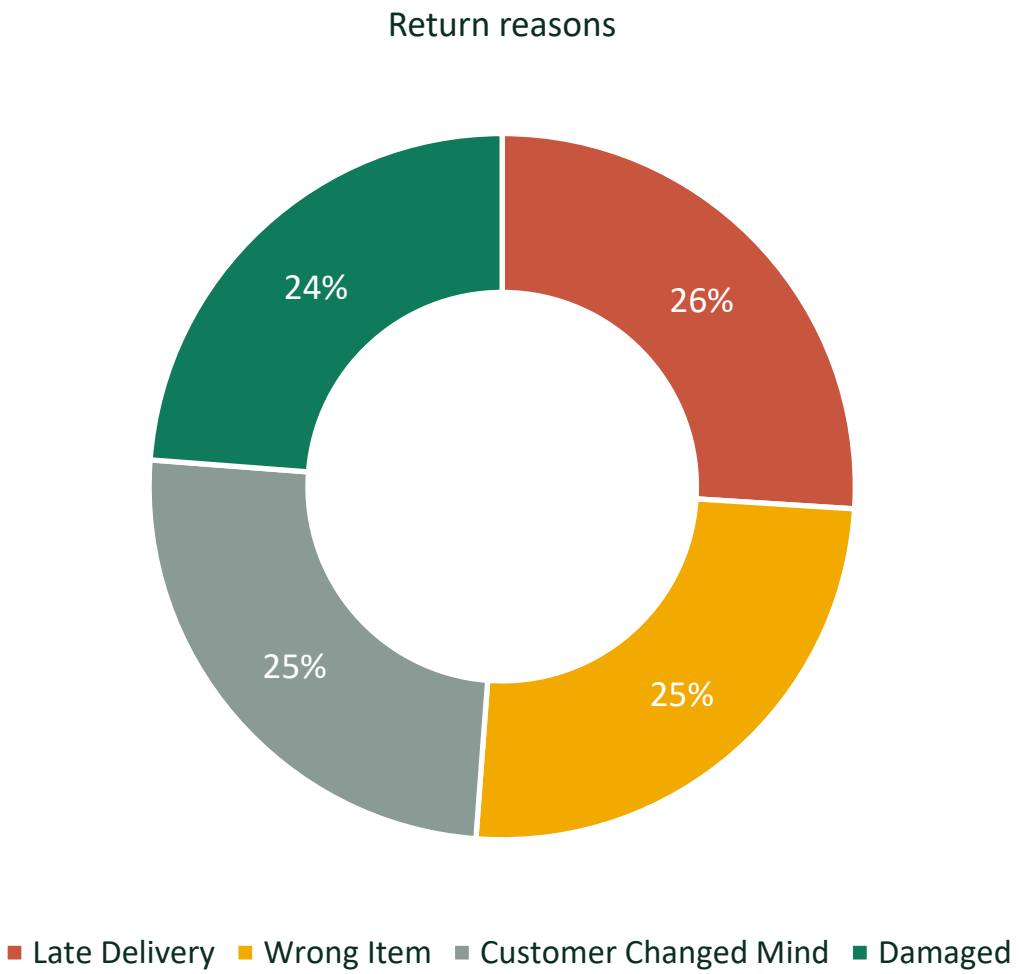
Gold, Silver and Bronze customers spend almost exactly the same per order (₦135k to ₦136k).

Gender and age groups show the same flat pattern.

So what?

The loyalty programme is not lifting order value. Test targeted offers instead of broad tiers.

About 75% of returns are fulfilment problems

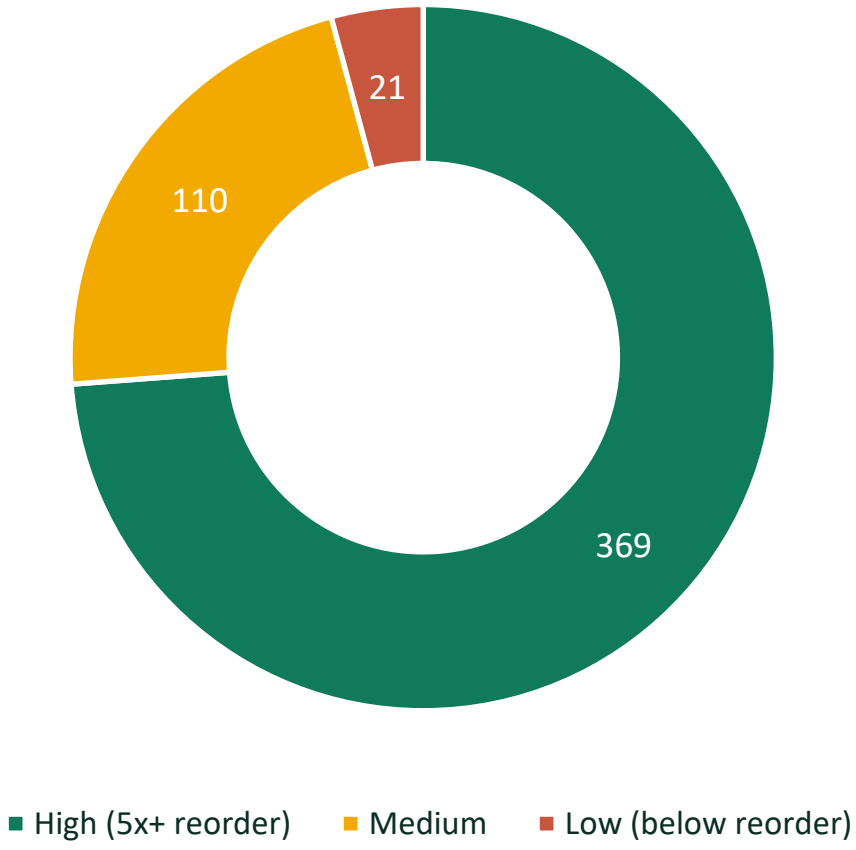




Late delivery, wrong item and damaged goods drive three in four returns. The rate is flat across channel, payment method and discount level, so the fix is operational.

Overstocked overall, with 21 critical gaps

Product-store lines by stock status



₦3.17bn

Stock value (1.4× annual revenue)

21

Lines below reorder level

5

Employees in every store

9 of 25

Stores with no Store Manager

Largest gaps: Home Item 302 (STR013) and Electronics Item 165 (STR014).

Data quality: clean on the surface, issues underneath



67% of returns sit on Pending or Cancelled orders

Logically impossible: a return needs a completed sale



About 5,000 orders predate the customer's registration

Flagged, not deleted



Store names repeat; Region conflicts with State

Group by Store_ID and use State



Order prices never match catalogue prices

Margin analysis is unreliable

0

missing values, exact duplicates or broken keys across 8 tables

Five recommendations

1	**Fix the order pipeline**	Follow up Pending orders and investigate cancellations
2	**Invest in Beauty and Fashion**	Range, promotion and stock depth; review Home
3	**Cut returns at the source**	Audit delivery times, picking accuracy and packaging
4	**Restock 21 lines, trim overstock**	Add automatic reorder alerts
5	**Improve data quality**	Returns links, unique store names, region mapping, delivery dates

Next: machine learning opportunities



Sales forecasting

Monthly, store and category series



Customer segmentation

RFM K-Means prototype built



Order completion

Predict which orders convert



Return prediction

Weak signal in current fields



Inventory forecasting

Reorder timing by product-store



Churn prediction

Recency from last purchase

Caveat: one year of data, and K-Means clusters are only moderately separated (silhouette about 0.3).

Thank you



The biggest opportunity is hiding in plain sight: converting the two thirds of orders that never complete.

Questions?